

级数

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1. Fourier 级数

傅里叶级数 $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l})$ $l = \frac{T}{2}$... 系 $1 \cos \frac{n\pi x}{l} \sin \frac{n\pi x}{l}$

傅里叶级数 系数满足

① $T = 2l$ $[-l, l]$ 上可积

$$\begin{cases} b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx \\ a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx \\ a_0 = \frac{1}{l} \int_{-l}^l f(x) dx \end{cases}$$

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l})$$

↓ Dirichlet

$$= \dots \text{且 } S(x) = \begin{cases} f(x) \\ \frac{f(x-0) + f(x+0)}{2} \end{cases}$$

② $f(x)$ $x \in [-l, l]$ 周期延拓 $F(x) = \begin{cases} f(x) & E(l, l) \\ f(x-k \cdot 2l) & \end{cases}$

③ $f(x)$ $x \in [0, l]$ 奇偶延拓 $F(x) = \begin{cases} f(x) & [0, l) \\ ? & (l, 0) \end{cases}$

$$F(x) = \begin{cases} f(x) & (0, l) \\ 0 & (l, 0) \\ -f(x) & E(l, 0) \end{cases}$$

$$F(x) = \begin{cases} f(x) & [0, l) \\ f(x) & E(l, 0) \end{cases}$$

延拓 → 展开 → 求和

记住

$$\begin{cases} \text{几何/等比} & \sum_{n=1}^{\infty} aq^{n-1} (a \neq 0) < \begin{cases} |q| < 1 \text{ 收} \\ |q| > 1 \text{ 发} \end{cases} \\ \text{p级数} & \sum_{n=1}^{\infty} \frac{1}{n^p} < \begin{cases} p \leq 1 \text{ 发} \\ p > 1 \text{ 收} \end{cases} \\ \text{调和} & \sum_{n=1}^{\infty} \frac{1}{n} \text{ 发} \end{cases}$$

2. 幂级数

$\sum_{n=0}^{\infty} a_n (x-x_0)^n$ 半径 R 区间 (x_0-R, x_0+R) 域 $E(R, R)$

判定 ① $\sum_{n=0}^{\infty} a_n x^n$ $\begin{cases} x=x_0 \text{ 收} \\ x \neq x_0 \text{ 发} \end{cases}$ $\begin{cases} |x| < |x_0| \text{ 绝收} \\ |x| > |x_0| \text{ 发} \end{cases}$

② $\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = R$ $\begin{cases} 0 < R < +\infty & (-R, R) \text{ 绝收} \\ R=0 & \text{仅 } x=0 \text{ 收} \\ R=+\infty & (-\infty, +\infty) \text{ 绝收} \end{cases}$ $\begin{cases} |x| < R \text{ 收} \\ |x| > R \text{ 发} \end{cases}$

③ $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = R$

本质

比较

单调性

$$\lim_{n \rightarrow \infty} a_n = 0$$

概念区分

部分和序列 S_k k 有限

$n \rightarrow \infty$ ↓ 级数 $\sum_{n=1}^{\infty} a_n$ $\sum_{n=1}^{\infty} (u_n(x))$

求 $\lim$ ↓ 和 - 和函数

应用 求 $f^{(n)}(x) = n! a_n$

/ $\lim$ / 不定积分

/ 定积分 (级项级数)

/ 推 Euler 公式

$e^{ix} = \cos x + i \sin x$

性质

$S_n = \sum_{k=0}^n a_k x^k$ 分子积

$\int_0^x S_n dx = \sum_{k=0}^n \frac{a_k}{k+1} x^{k+1}$ 分母导

$S_n^{(k)} = n! a_n + (n+1)! a_{n+1} x + \dots \geq a_{n+1} x + \dots$

端点: $\sum a_n x^n$ 收 + $S(x)$ 在 R 处 $\begin{cases} \text{连续} \\ \text{不连续} \end{cases}$

求和

$\sum_{n=1}^{\infty} \frac{x^n}{n} = -\ln(1-x) \quad (-1, 1)$ $\sum_{n=1}^{\infty} n x^n = \frac{x}{(1-x)^2}$

$\sum_{n=1}^{\infty} \frac{x^{n+1}}{n} = -x \ln(1-x)$

$\sum_{n=1}^{\infty} \frac{x^{n+1}}{n} = \begin{cases} -\frac{\ln(1-x)}{x} & (-1, 0) \cup (0, 1) \\ 1 & x=0 \end{cases}$

$\sum_{n=1}^{\infty} \frac{x^n}{n(n+1)} = \begin{cases} \frac{1}{x} \ln(1-x) + 1 & (-1, 0) \cup (0, 1) \\ 0 & x=0 \\ 1 & x=1 \end{cases}$

函数展开成幂级数

$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$

$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$

$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} \quad (-1, 1)$

$(1+x)^a = 1 + \sum_{n=1}^{\infty} \frac{a \dots (a-n+1)}{n!} x^n \quad (-1, 1)$

$\frac{1}{1-x} = \sum_{n=0}^{\infty} (1)^n x^n$

$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$

$\frac{1}{1-x^2} = \sum_{n=0}^{\infty} (1)^n x^{2n}$

$\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} \quad (-1, 1)$

$\frac{1}{\sqrt{1-x^2}} = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n (2n-1)!!}{(2n)!!} x^{2n} \quad (-1, 1)$

$\arcsin x = x + \sum_{n=1}^{\infty} \frac{(2n-1)!!}{(2n)!!} \frac{1}{2n+1} x^{2n+1} \quad (-1, 1)$



CS 扫描全能王

3亿人都在用的扫描App

3. 敛散性证明 (条/绝)

① 性质 $\sum (k a_n + b_n) \checkmark \quad \sum a_n \checkmark + b_n \checkmark \Rightarrow \sum (a_n + b_n) \checkmark$

② 正项 $\exists M \forall n \in \mathbb{N} \quad S_n \leq M$

$$a_n \leq b_n \dots \Rightarrow \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c \quad \begin{cases} 0 < c < +\infty \\ c = 0 \\ c = +\infty \end{cases}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \rho \quad \begin{cases} \rho < 1 \\ \rho > 1 \\ \rho = 1 \end{cases}$$

$f(x)$ 在 $[1, +\infty)$ 非负 $\downarrow \Rightarrow \sum_{n=1}^{\infty} a_n$ 与 $\int_1^{+\infty} f(x) dx$ 同 $\checkmark$

③ 一般 Leibniz: $u_n \downarrow \rightarrow 0 \Rightarrow \sum (-1)^n u_n \checkmark \quad S \leq u_1$ 条件 (带)

$\sum_{n=1}^{\infty} |u_n| \checkmark \Rightarrow \sum u_n \checkmark$ 绝发/逆 $\times$

$$\lim_{n \rightarrow \infty} \frac{|u_{n+1}|}{|u_n|} = \rho \quad \left/ \quad \lim_{n \rightarrow \infty} \sqrt[n]{|u_n|} = \rho \right. \quad \begin{cases} \rho < 1 \\ \rho > 1 \\ \rho = 1 \end{cases} \quad \text{绝/发 (2种)}$$

泰勒级数

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n \quad \text{充要}$$

$$\Leftrightarrow |x-x_0| < R \text{ 内 } \lim_{n \rightarrow \infty} R_n(x) = \lim_{n \rightarrow \infty} \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)^{n+1} = 0$$

$$\text{麦克劳林级数} \quad \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x-x_0)^n$$



空间解几

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1. 空间解几

向量定理 $\exists$ 唯一 $\vec{a} = m\vec{b}$ / $\vec{a} = m_1\vec{b} + m_2\vec{c}$ / $\vec{a} = m_1\vec{e}_1 + m_2\vec{e}_2 + m_3\vec{e}_3$

运算

$$\begin{aligned} ① \vec{a} \cdot \vec{b} &= 0 \Leftrightarrow \vec{a} \perp \vec{b} & |\vec{a}| &= \sqrt{\vec{a} \cdot \vec{a}} \\ ② \vec{a} \times \vec{b} &= 0 \Leftrightarrow \vec{a} \parallel \vec{b} \Leftrightarrow \frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} \\ ③ (\vec{a} \times \vec{b}) \cdot \vec{c} &= \vec{a} \cdot (\vec{b} \times \vec{c}) \Leftrightarrow \vec{a}, \vec{b}, \vec{c} \text{ 共面} \Leftrightarrow \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0 \\ ④ \vec{a} \times (\vec{b} \times \vec{c}) &= (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} \end{aligned}$$

$$\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$$

$$\vec{a} \cdot [(\vec{b} + \vec{d}) \times \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) + \vec{a} \cdot (\vec{d} \times \vec{c})$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{b} \cdot (\vec{a} \times \vec{c}) = \vec{c} \cdot (\vec{a} \times \vec{b})$$

应用 $l = (\vec{AB})_{\vec{AC}} = \vec{AB} \cdot \vec{AC}$

$$S_{\Delta} = \frac{1}{2} |\vec{AB} \times \vec{AC}|$$

$$V = \frac{1}{6} |(\vec{AB} \times \vec{AC}) \cdot \vec{AD}|$$

$$\vec{n} \cdot \vec{p}_0 \vec{p} = 0$$

平面方程 $ACx + Bcy + Cz + D = 0$ / $Ax + By + Cz + D = 0$ $\vec{n} = A\vec{i} + B\vec{j} + C\vec{k}$ / $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ $\vec{a} = a\vec{i} + b\vec{j} + c\vec{k}$

角 $\cos \langle \vec{v}_1, \vec{v}_2 \rangle = \frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}}$ // $\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$ $\perp: A_1A_2 + B_1B_2 + C_1C_2 = 0$

距离 $P(x_0, y_0, z_0)$ 到 $Ax + By + Cz + D = 0$ $d = |\vec{PP}_0| |\cos \langle \vec{PP}_0, \vec{n} \rangle| = |\vec{PP}_0 \cdot \vec{e}_n| = \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}$

直线方程 $\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$ / $\begin{cases} x = x_0 + at \\ y = y_0 + bt \\ z = z_0 + ct \end{cases} t \in (-\infty, +\infty)$ / $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$ / $\begin{cases} A_1x + B_1y + C_1z + D_1 = 0 \\ A_2x + B_2y + C_2z + D_2 = 0 \end{cases}$

平面束 $\lambda(A_1x + B_1y + C_1z + D_1) + \mu(A_2x + B_2y + C_2z + D_2) = 0$

角 $\cos \langle \vec{v}_1, \vec{v}_2 \rangle = \frac{l_1l_2 + m_1m_2 + n_1n_2}{\sqrt{l_1^2 + m_1^2 + n_1^2} \sqrt{l_2^2 + m_2^2 + n_2^2}}$ / $\frac{\pi}{2} - \cos \langle \vec{v}, \vec{n} \rangle < \begin{cases} \frac{\pi}{2} & \vec{v} \perp \vec{n} \\ \frac{\pi}{2} & \vec{v} \parallel \vec{n} \end{cases}$ $\frac{A}{a} = \frac{B}{b} = \frac{C}{c}$ $Ax + By + Cz + D = 0$

距离 点线 $d = \frac{|\vec{MP} \times \vec{v}|}{|\vec{v}|}$ / 作垂面 $d = |\vec{AB}|$

线线 L_1 上 t , L_2 上 s 二元函数最值 / $\vec{v} = \vec{v}_1 \times \vec{v}_2$ $d = \frac{|\vec{M}_1 \cdot \vec{M}_2 \vec{v}|}{|\vec{v}|}$

投影 $\vec{n} = \vec{n}_1 \times \vec{n}_2$ 过 L 作 $\pi_1 \perp \pi \Rightarrow \{ \pi, \pi_1 \}$ 交线

2. 场的方向导数与阶梯

场 $u = u(P) = u(x, y, z)$ $\vec{A} = A_x(P)\vec{i} + A_y(P)\vec{j} + A_z(P)\vec{k}$

方向 l 上单位向量

$$\vec{n}_0 = \frac{\vec{n}}{|\vec{n}|} = (\cos \alpha, \cos \beta, \cos \gamma)$$

方向导数 $\begin{aligned} u(P) \text{ 在 } P_0 \text{ 处沿 } l \frac{\partial u}{\partial l} \Big|_{P_0} &= \lim_{P \rightarrow P_0} \frac{u(P) - u(P_0)}{|\vec{PP}_0|} = \lim_{P \rightarrow P_0} \frac{u(P) - u(P_0)}{\rho} = \lim_{P \rightarrow P_0} \frac{\Delta u}{\rho} \\ &= \lim_{\rho \rightarrow 0} \frac{f(x_0 + \rho \cos \alpha, y_0 + \rho \cos \beta, z_0 + \rho \cos \gamma) - f(x_0, y_0, z_0)}{\rho} \quad f(x, y, z) \text{ 在 } (x_0, y_0, z_0) \text{ 可微} \rightarrow \frac{\partial f}{\partial x} \Big|_{P_0} = f'_x(x_0, y_0, z_0) \cos \alpha + f'_y(x_0, y_0, z_0) \cos \beta + f'_z(x_0, y_0, z_0) \cos \gamma \\ &= u(x, y, z) \text{ 在 } (x_0, y_0, z_0) \text{ 可微} \rightarrow \frac{\partial u}{\partial l} \Big|_{P_0} = \frac{\partial u}{\partial x} \Big|_{P_0} \cos \alpha + \frac{\partial u}{\partial y} \Big|_{P_0} \cos \beta + \frac{\partial u}{\partial z} \Big|_{P_0} \cos \gamma \end{aligned}$

梯度 $\text{grad} u \Big|_{P_0} = \left(\frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} + \frac{\partial u}{\partial z} \vec{k} \right) \Big|_{P_0}$

$$\leftarrow = \left| \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right) \right|_{P_0} (\cos \alpha, \cos \beta, \cos \gamma)$$

数量 $\frac{\partial u}{\partial l} \Big|_{P_0} = \text{grad} u \Big|_{P_0} \cdot \vec{l}^0 \leq |\text{grad} u \Big|_{P_0}|$

法则 $\text{grad} (\alpha u + \beta v) = \alpha \text{grad} u + \beta \text{grad} v$

$$\text{grad} (u \cdot v) = u \text{grad} v + v \text{grad} u$$

$$\text{grad} f(u) = f'(u) \text{grad} u$$



微分应用

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10. 多元函数极值

判定 $\begin{cases} \text{必要 } f'_x(x_0, y_0) = 0, f'_y(x_0, y_0) = 0 \\ \text{充分 邻域内 } f, f', f'' \text{ 连续} \end{cases}$

驻点/无偏导点

若 $f'_x(x_0, y_0) = 0, f'_y(x_0, y_0) = 0$

令 $f''_{xx}(x_0, y_0) = A, f''_{xy}(x_0, y_0) = B, f''_{yy}(x_0, y_0) = C$

$\begin{cases} B^2 - AC < 0 \checkmark \\ B^2 - AC > 0 \times \\ B^2 - AC = 0 ? \end{cases}$

求法 条件极值: $z = f(x, y)$ 在 $\varphi(x, y) = 0$ 下

不满足 ① Lagrange 函数 $F(x, y, \lambda) = f(x, y) + \lambda \varphi(x, y)$

满足 ② 隐函数 $\frac{dz}{dx} = 0 \leftarrow \varphi(x, y) = 0$

$\begin{cases} F'_x = 0 \\ F'_y = 0 \\ F'_\lambda = \varphi(x, y) = 0 \end{cases}$

一般极值

① $\begin{cases} f'_x = 0 \\ f'_y = 0 \end{cases}$ 求所有嫌疑点

② 分别求 $B^2 - AC$

应用 向量函数 $|\vec{v}| = \left| \frac{d\vec{r}}{dt} \right| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$

曲线的切线 $\frac{x - x_0}{x'(t_0)} = \frac{y - y_0}{y'(t_0)} = \frac{z - z_0}{z'(t_0)}$

法平面 $x'(t_0)(x - x_0) + y'(t_0)(y - y_0) + z'(t_0)(z - z_0) = 0$

切向量 $\vec{T} = \left\{ \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\} \Big|_{t=t_0} = \{dx, dy, dz\} = \left\{ \frac{dx}{ds}, \frac{dy}{ds}, \frac{dz}{ds} \right\}$

曲面的法线 $\frac{x - x_0}{F'_x|_M} = \frac{y - y_0}{F'_y|_M} = \frac{z - z_0}{F'_z|_M}$

切平面 $F'_x|_M(x - x_0) + F'_y|_M(y - y_0) + F'_z|_M(z - z_0) = 0$

法线 $\vec{n} = \{F'_x, F'_y, F'_z\}$

$z = f(x, y) \Rightarrow \{f'_x, f'_y, -1\}$

$F(x, y, z) = f(x, y) - z$

$f(x_0 + h, y_0 + k)$

泰勒公式 (f 在 P_0 处的 n 阶) 有 $n+1$ 阶连续偏导

对 $U(P_0)$ 内 $\forall (x_0 + h, y_0 + k) \exists \theta \in (0, 1)$ 使 $f(x_0 + h, y_0 + k) = f(x_0, y_0) + h \frac{\partial f}{\partial x} + k \frac{\partial f}{\partial y} + \frac{1}{2!} (h \frac{\partial^2}{\partial x^2} + k \frac{\partial^2}{\partial y^2}) f(x_0, y_0) + \dots + \frac{1}{(n+1)!} (h \frac{\partial^2}{\partial x^2} + k \frac{\partial^2}{\partial y^2})^{n+1} f(x_0, y_0)$

11. 二重积分 - 计算

定义 $\iint_D f(x, y) d\sigma = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i, \eta_i) \Delta \sigma_i$ 连续 $\rightarrow$ 可积 $\rightarrow$ 有界

性质 $|\iint_D f(x, y) d\sigma| \leq \iint_D |f(x, y)| d\sigma, m \leq \iint_D f(x, y) d\sigma \leq M$

$\exists (\xi, \eta) \in D, \iint_D f(x, y) d\sigma = f(\xi, \eta) \sigma$

计算 平面 $\int_a^b dx \int_{\varphi(x)}^{\psi(x)} f(x, y) dy$ 对称 $\begin{cases} b-x \text{ 轴} < f(x, y) - y \text{ 奇} \\ \text{偶} \end{cases}$

极 $\iint_D f(\rho \cos \theta, \rho \sin \theta) \rho d\rho d\theta$ 极点 O 在 D 内/边/外

一般 $\iint_D f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$ $\begin{cases} x = a \cos \theta \\ y = b \sin \theta \end{cases} d\sigma = \left| \frac{\partial(x, y)}{\partial(r, \theta)} \right| dr d\theta = \left| \frac{\partial x}{\partial r} \frac{\partial y}{\partial \theta} - \frac{\partial x}{\partial \theta} \frac{\partial y}{\partial r} \right| dr d\theta = ab r dr d\theta$

应用 $V = \iint_D (z_{\text{上}} - z_{\text{下}}) d\sigma$ 曲顶柱体

$M = \iint_D \mu(x, y) d\sigma$ 薄板质量

$\bar{x} = \frac{\iint_D x \mu(x, y) d\sigma}{\iint_D \mu(x, y) d\sigma}$

$\bar{y} = \frac{\iint_D y \mu(x, y) d\sigma}{\iint_D \mu(x, y) d\sigma}$

$ds = \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \frac{dx dy}{d\sigma}$

$= \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2} dt$

$= \sqrt{r(\theta)^2 + r'(\theta)^2} d\theta$

积分汇总



12. 三重积分 - 应用

定义 $\iiint_V f(x,y,z) dv = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i, \eta_i, \zeta_i) \Delta V_i$ 立体的质量

计算：投影 $V = \{(x,y,z) | z_1(x,y) \leq z \leq z_2(x,y), (x,y) \in D_{xy}\}$

① 平面
+ 截面
对称

$$\iiint_V f(x,y,z) dv = \iint_{D_{xy}} \left[\int_{z_1(x,y)}^{z_2(x,y)} f(x,y,z) dz \right] d\sigma = \int_a^b dx \int_{y_1(x)}^{y_2(x)} \int_{z_1(x,y)}^{z_2(x,y)} f(x,y,z) dz dy$$

$$f(x,y,z) = \rho(x,y,z)$$

$$\iiint_V \rho(x,y,z) dv = \int_{x_1}^{x_2} \rho(x) A(x) dx$$

② 柱面

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases} \quad \iiint_V f(x,y,z) dv = \int_0^{2\pi} d\theta \int_{r_1(\theta)}^{r_2(\theta)} r dr \int_{z_1(r \cos \theta, r \sin \theta)}^{z_2(r \cos \theta, r \sin \theta)} dz = \iiint_V f(r \cos \theta, r \sin \theta, z) r dr d\theta dz$$

求 J 体/板 $I_L = \iiint_V \rho r^2 dx dy dz$

③ 球

$$\begin{cases} x = \rho \sin \varphi \cos \theta \\ y = \rho \sin \varphi \sin \theta \\ z = \rho \cos \varphi \end{cases} \quad dv = d\rho \cdot (\rho \sin \varphi d\theta) \cdot \rho d\varphi = \rho^2 \sin \varphi d\rho d\theta d\varphi$$

$$\iiint_V f(x,y,z) dv = \iiint_V f(\rho \sin \varphi \cos \theta, \rho \sin \varphi \sin \theta, \rho \cos \varphi) \rho^2 \sin \varphi d\rho d\theta d\varphi$$

求 F $F_x = G \iiint_V \frac{x-x_0}{r^3} \rho(x,y,z) dv$

④ 一般 $\iiint_V f(x,y,z) dv = \iiint_V [f(x(u,v,w), y(u,v,w), z(u,v,w))] abc \rho^2 \sin \varphi du dv dw$

二次曲面 - 积分上下限边界

球 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ $\left\langle \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \right\rangle$ 球面

锥 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$

抛物面 $\begin{cases} \frac{x^2}{a^2} + \frac{y^2}{b^2} = \pm 2z \text{ 椭圆} \\ \frac{x^2}{a^2} - \frac{y^2}{b^2} = 2z \text{ 双曲} \end{cases}$

双曲 $\begin{cases} \text{单叶} \frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \\ \text{双叶} \frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = -1 \end{cases}$

柱 $\begin{cases} \text{椭圆} \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \\ \text{双} \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \\ \text{抛物} x^2 = 2ay \end{cases}$

路径无关① 或 闭曲线 $\oint P dx + Q dy = 0$

② 或 闭曲线 $\oint_{AB} P dx + Q dy$ 与 A, B 有关

③ D 内 $\exists u(x,y)$ 使 $du = P dx + Q dy$

④ $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

循环常数

13. 第二类曲线积分

定义 $\int_{\Gamma} \vec{A} \cdot \vec{T} dl = \int_{\Gamma} \vec{A} \cdot d\vec{r}$

$$= \int_{\Gamma} [P(x,y,z) \cos \alpha + Q(x,y,z) \cos \beta + R(x,y,z) \cos \gamma] dl$$

$$= \int_{\Gamma} [P(x,y,z) dx + Q(x,y,z) dy + R(x,y,z) dz]$$

性质 $\int_{\Gamma^+} (\vec{A} \cdot \vec{e}_r) dl = - \int_{\Gamma^-} (\vec{A} \cdot \vec{e}_r) dl$

$$\int_{\Gamma} (\vec{A} \cdot \vec{e}_r) dl = \int_{\Gamma^+} (\vec{A} \cdot \vec{e}_r) dl + \int_{\Gamma^-} (\vec{A} \cdot \vec{e}_r) dl$$

计算 $\int_{\Gamma_{AB}} \vec{A} \cdot \vec{e}_r ds = \int_{t_A}^{t_B} [P x'(t) + Q y'(t) + R z'(t)] dt$

变一元函数定积分

格林 $\Gamma/P/Q$ 连一阶导

$$\oint_{\Gamma} P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) d\sigma = \iint_D \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ P & Q \end{vmatrix} d\sigma$$

变二重积分

原函数

$$u(x,y) = \int_{(x_0,y_0)}^{(x,y)} P(x,y) dx + Q(x,y) dy + C \xrightarrow{(0,0) \in D} \int_0^x P(x,0) dx + \int_0^y Q(x,y) dy + C$$

N-L 公式

$$\int_{(x_0,y_0)}^{(x_1,y_1)} P dx + Q dy = u(x_1,y_1) - u(x_0,y_0)$$

变一元定积分



微分汇总

极限/连续/可微

极限 $\forall \varepsilon > 0, \exists \delta > 0$, 当 $P \in U(P_0, \delta)$ 时, 有 $|f(P) - A| < \varepsilon$, 则 $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = A$ 多元不可洛

内点
外点
边界点

$\Delta \lim_{x \rightarrow x_0} \lim_{y \rightarrow y_0} f(x,y), \lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} f(x,y), \lim_{x \rightarrow x_0} f(x,y)$ 都 $\exists$, 则三者等

可微 全增量 $\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$

(必要: ① $f'_x(x,y), f'_y(x,y) \exists$ ② $z = f(x,y)$ 在 (x,y) 处连续
- 证: $\Delta y = A \Delta x + o(\Delta x) (\Delta x \rightarrow 0) \quad dy = f'_y dx$

连续 $\left\{ \begin{array}{l} \text{开: } \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \Delta z = 0 \\ \text{闭: } \forall P_0 \in D \quad \lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(x,y) = f(x_0, y_0) \end{array} \right.$

证不可微:

① $f'_x(x_0, y_0), f'_y(x_0, y_0)$ 是否都 $\exists$

② $\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\Delta z - f'_x(x_0, y_0) \Delta x - f'_y(x_0, y_0) \Delta y}{\rho} \neq 0$
 $\downarrow$
 充要

偏导数/全微分

偏导 $f'_x(x_0, y_0) = \lim_{\Delta x \rightarrow 0} \frac{\Delta_x z}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} = \lim_{x \rightarrow x_0} \frac{f(x, y_0) - f(x_0, y_0)}{x - x_0}$ 混合连续 $\rightarrow$ 顺序无关

全微分 $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$

全增量 $\Delta z = f'_x(x_0, y_0) \Delta x + f'_y(x_0, y_0) \Delta y + \varepsilon_1 \Delta x + \varepsilon_2 \Delta y \quad \lim \varepsilon_1 = 0 \quad \lim \varepsilon_2 = 0$ 近似 $\Delta z \approx dz$

复合函数 $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} \quad \frac{dz}{dx} = \frac{\partial z}{\partial u} \frac{du}{dx} + \frac{\partial z}{\partial v} \frac{dv}{dx}$

$z = f[u(x,y), v(y)] \quad z < \sum_{i=1}^n \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} \quad \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x}$

形式不变 $dz = \frac{\partial z}{\partial s} ds + \frac{\partial z}{\partial t} dt \quad \xrightarrow{x=x(s,t) \quad y=y(s,t)} dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$

隐函数

存在定理: ① $F(x_0, y_0, z_0) = 0$ ② $P(x_0, y_0, z_0)$ 邻域内连续 F'_x, F'_y ③ $F'_z(x_0, y_0, z_0) \neq 0$

则能唯一确定 $z = f(x,y)$ 连续 & 偏导连续, $z_0 = f(x_0, y_0) \quad \frac{\partial z}{\partial x} = -\frac{F'_x}{F'_z} \quad \frac{\partial z}{\partial y} = -\frac{F'_y}{F'_z}$

隐函数组求: $\begin{cases} F(x,y,u,v) = 0 \\ G(x,y,u,v) = 0 \end{cases} \Rightarrow \begin{cases} u = u(x,y) \\ v = v(x,y) \end{cases}$ ① 地位同, 微分 ② 分别 ∂_x, ∂_y

场的方向导数

方向导数 $\frac{\partial f}{\partial l} \Big|_{P_0} = \lim_{\rho \rightarrow 0} \frac{f(P) - f(P_0)}{\rho} = \lim_{\rho \rightarrow 0} \frac{f(\rho \cos \alpha, \rho \sin \alpha) - f(x_0, y_0)}{\rho}$

梯度 $\frac{\partial f}{\partial l} \Big|_{P_0} = \text{grad} u \Big|_{P_0} \cdot \vec{l}^0 \leq |\text{grad} u \Big|_{P_0}| = \left| \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \right|$

空间曲线

旋转曲面: 曲线 Γ 绕 OY 轴旋转得

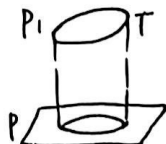
$\forall M(x,y,z)$ 设 M 由 $M_1(x_1, y_1, z_1) \in \Gamma$ 转得

$\begin{cases} z_1 = \sqrt{y_1} \\ x_1 = 0 \end{cases} \xrightarrow{\text{代入}} \begin{cases} y = y_1 \\ \sqrt{x^2 + z^2} = |z_1| \end{cases}$

投影曲线

① oxy 上 = 消 z

② 柱面与 π 交线: $\forall P(x,y,z)$ 柱面上任一点
 使 $\vec{PP} \parallel \vec{n}$



积分汇总

名称

定义+性质

计算

二重积分

$$\iint_D f(x,y) d\sigma = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i, \eta_i) \Delta \sigma_i$$

线性/折分 $1 \leq i \leq m$ / $1 \leq j \leq n$

$$f(x,y) \text{ 在 } D \text{ 上连续, } \exists (\xi, \eta) \in D, \iint_D f(x,y) d\sigma = f(\xi, \eta) \Delta \sigma$$

应用

曲顶柱体 $V = \iint_D z(x,y) d\sigma$

平面薄板 $M = \iint_D \mu(x,y) d\sigma$

$$\bar{x} = \frac{\iint_D x d\sigma}{D}, \bar{y} = \frac{\iint_D y d\sigma}{D}$$

$$F_z = -\mu k m h \iint_D \frac{1}{(x^2+y^2+h^2)^{\frac{3}{2}}} d\sigma$$

$$J = \iint_D (x^2+y^2) \mu(x,y) d\sigma$$

① 平面

$$\iint_D f(x,y) d\sigma = \int_a^b dx \int_{\phi(x)}^{\psi(x)} f(x,y) dy$$

$$= \int_c^d dy \int_{\phi(y)}^{\psi(y)} f(x,y) dx$$

② 极

$$\iint_D f(x,y) d\sigma = \iint_D f(r \cos \theta, r \sin \theta) r dr d\theta$$

③ 一般

$$\iint_D f(x,y) d\sigma = \iint_D f(x(u,v), y(u,v)) \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du dv$$

$$\begin{cases} x = a r \cos \theta \\ y = b r \sin \theta \end{cases}$$

$$= \iint_D f(x,y) a b r dr d\theta$$

看①在哪个轴

θ型 $\begin{cases} \text{外 } r_1(\theta) \leq r \leq r_2(\theta) & \alpha \leq \theta \leq \beta \\ \text{上 } 0 \leq r \leq r_1(\theta) & \alpha \leq \theta \leq \beta \\ \text{内 } 0 \leq r \leq r_1(\theta) & 0 \leq \theta \leq 2\pi \end{cases}$

r型 $\begin{cases} x^2+y^2=R^2 & [0, 2\pi] [0, R] \\ x^2+y^2=2xR & [-\frac{\pi}{2}, \frac{\pi}{2}] [0, 2R \cos \theta] \\ x^2+y^2=2yR & [0, \pi] [0, 2R \sin \theta] \end{cases}$

三重积分

应用

$$V = \iiint_D dV$$

$$M = \iiint_D \mu(x,y,z) dV$$

$$\bar{x} = \frac{\iiint_D x dV}{V}, \bar{y} = \frac{\iiint_D y dV}{V}, \bar{z} = \frac{\iiint_D z dV}{V}$$

$$F_z = \iiint_D \frac{k \mu r dr d\theta dz}{r^2+z^2} = k \mu m \iint_D \frac{z}{(x^2+y^2+z^2)^{\frac{3}{2}}} d\sigma$$

$$J = \iiint_D (x^2+y^2) \mu(x,y,z) dV$$

$$\iiint_D f(x,y,z) dV = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i, \eta_i, \zeta_i) \Delta V_i$$

① 投影 表二重积分 (平行轴与边界曲面交点)

$$\iiint_D f(x,y,z) dV = \iint_{D_{xy}} \left[\int_{z_1(x,y)}^{z_2(x,y)} f(x,y,z) dz \right] d\sigma = \int_a^b dx \int_{\phi(x)}^{\psi(x)} dy \int_{z_1(x,y)}^{z_2(x,y)} f dz$$

Dxy用表

(f(x,y,z)反表达式及Dz易算) ② 截面 一元函数积分+二重积分 $V = \int_{z_1}^{z_2} dV = \int_{z_1}^{z_2} D_z dz$

$$\iiint_D f(z) dV = \int_{z_1}^{z_2} f(z) A(z) dz = \int_{z_1}^{z_2} f(z) \iint_{D_z} d\sigma dz = \int_{z_1}^{z_2} f(z) d\sigma dz$$

(投影为圆一部分) ③ 柱面

$$\iiint_D f(x,y,z) dV = \iint_{D_{xy}} d\sigma \left(\int_{z_1}^{z_2} f(x,y,z) dz \right) = \int_{\alpha}^{\beta} d\theta \int_{r_1(\theta)}^{r_2(\theta)} r dr \int_{z_1(r \cos \theta, r \sin \theta)}^{z_2(r \cos \theta, r \sin \theta)} f dz$$

④ 球

$$\iiint_D f(x,y,z) dV = \iiint_D f(\rho \sin \varphi \cos \theta, \rho \sin \varphi \sin \theta, \rho \cos \varphi) \rho^2 \sin \varphi d\rho d\varphi d\theta$$

$$\begin{cases} x = \rho \sin \varphi \cos \theta \\ y = \rho \sin \varphi \sin \theta \\ z = \rho \cos \varphi \end{cases}$$

⑤ 一般

$$\iiint_D f(x,y,z) dV = \iiint_D f(x(u,v,w), y(u,v,w), z(u,v,w)) \left| \frac{\partial(x,y,z)}{\partial(u,v,w)} \right| du dv dw$$

第一类曲线积分

→ 元定

$$\int_C f(x,y,z) ds = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i, \eta_i, \zeta_i) \Delta s_i = \int_a^b f(x(t), y(t), z(t)) \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2} dt = \int_a^b f(x(t), y(t), z(t)) \sqrt{1 + y'(x)^2 + z'(x)^2} dx = \int_{\alpha}^{\beta} f(r \cos \theta, r \sin \theta) \sqrt{r^2 + r'^2} d\theta$$

第二类曲线积分

→ 重积分2

$$\int_C f(x,y,z) ds = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i, \eta_i, \zeta_i) \Delta s_i = \iint_{D_{xy}} f(x,y,z(x,y)) \sqrt{1+z'(x)^2+z'(y)^2} d\sigma$$

Summary $\begin{cases} [a,b] \\ T \subset R^2/R^3 \\ 0 \subset R^2 \\ S \subset R^3 \\ V \subset R^3 \end{cases}$

无函数定
第一类曲线 → 第二类曲线 R^2 格 R^3 新
→ 重积分
第一类曲面 → 第二类曲面 R^3 高
→ 三重积分

$$P(x,y,z(x,y)) \quad r = \pm \left\{ \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y} \right\}$$

$$\cos \varphi = \pm \frac{1}{\sqrt{1+z'(x)^2+z'(y)^2}} \quad S = \iint_{D_{xy}} \sqrt{1+z'(x)^2+z'(y)^2} d\sigma$$

特别地 $S = x^2+y^2+z^2=a^2$

$$\vec{n} = \pm \left\{ \frac{x}{a}, \frac{y}{a}, \frac{z}{a} \right\} \quad d\sigma = ds |\cos \varphi|$$

$$ds = \left| \frac{a}{z} \right| d\sigma = \frac{a}{\sqrt{a^2-x^2-y^2}} d\sigma$$



第二类曲线

二重积分

$$\int_C \vec{A} \cdot d\vec{l} = \int_C [P(x,y,z) \cos\alpha + Q(x,y,z) \cos\beta + R(x,y,z) \cos\gamma] dl$$

$$(3) = \int_C P(x,y,z) dx + Q(x,y,z) dy + R(x,y,z) dz = \int_C \vec{A} \cdot d\vec{l}$$

原函数 - 凑微分
循环常数
N-L公式

① 格林公式 D 有界闭 P, Q 连续 - 阶导 $\oint_C Pdx + Qdy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \iint_D \left| \frac{\partial P}{\partial x} \quad \frac{\partial Q}{\partial y} \right| dx dy$

路径无关 ① 先闭 $\oint_C Pdx + Qdy = 0$

② 先 $\int_{AB} Pdx + Qdy$ 无关

③ $\exists u(x,y), du = Pdx + Qdy \Leftrightarrow \frac{\partial u}{\partial x} = P \quad \frac{\partial u}{\partial y} = Q$

④ $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

第二类曲面

$$\iint_S \vec{A} \cdot \vec{n} ds = \iint_S [P(x,y,z) \cos\alpha + Q(x,y,z) \cos\beta + R(x,y,z) \cos\gamma] ds$$

$$= \iint_S [P(x,y,z) dy dz + Q(x,y,z) dx dz + R(x,y,z) dx dy] = \iint_S \vec{A} \cdot d\vec{S}$$

$\vec{A}(M) = P(x,y,z)\vec{i} + Q(x,y,z)\vec{j} + R(x,y,z)\vec{k}$
 $\vec{n} = \vec{e}_n(M) = \cos\alpha\vec{i} + \cos\beta\vec{j} + \cos\gamma\vec{k}$

$S: z = z(x,y)$

γ 为 S 法向量与 z 轴夹角 $\iint_S R(x,y,z) dx dy = \begin{cases} \iint_{D_{xy}} R(x,y,z(x,y)) dx dy & 0 \leq \gamma < \frac{\pi}{2} \\ 0 & \gamma = \frac{\pi}{2} \\ -\iint_{D_{xy}} R(x,y,z(x,y)) dx dy & \frac{\pi}{2} < \gamma \leq \pi \end{cases}$
① 二重积分 $\cos\gamma = \text{sgn}(\frac{\pi}{2} - \gamma) |\cos\gamma|$

② 三重积分 高斯公式 $\iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dx dy dz = \oiint_S P dy dz + Q dx dz + R dx dy = \oiint_S [P \cos\alpha + Q \cos\beta + R \cos\gamma] ds$

② 斯托克斯公式 (空间曲线)

$$\oint_L Pdx + Qdy + Rdz = \iint_S \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dy dz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) dz dx + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \iint_S \begin{vmatrix} dy dz & dz dx & dx dy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

路径无关 ① 先闭 $\oint_L Pdx + Qdy + Rdz = 0$

② 先 $\int_C Pdx + Qdy + Rdz$ 无关

③ $du = Pdx + Qdy + Rdz$

④ $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \quad \frac{\partial Q}{\partial z} = \frac{\partial R}{\partial y} \quad \frac{\partial P}{\partial z} = \frac{\partial R}{\partial x}$

$\oiint_S \vec{A} \cdot \vec{n} ds = \iiint_V \nabla \cdot \vec{A} dv = \iiint_V \text{div} \vec{A} dv$

$\oint_L \vec{A} \cdot \vec{T} dl = \iint_S \nabla \times \vec{A} ds = \iint_S \text{rot} \vec{A} ds$

环量与旋度 (空间曲线)

环量/环流量 $\oint_L \vec{A} \cdot d\vec{l}$

平均环流量/环量密度

环量密度 $\lim_{S \rightarrow 0} \frac{\oint_L \vec{A} \cdot d\vec{l}}{S} = \text{rot} \vec{A}$

旋度 (Stokes 条件) $\oint_L \vec{A} \cdot d\vec{l} = \iint_S \text{rot} \vec{A} \cdot d\vec{S}$ $\text{rot} \vec{A} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = \lim_{D \rightarrow 0} \frac{\oint_D \vec{A} \cdot d\vec{l}}{D}$

$\vec{A}_M$ 在 M 点绕 $\text{rot} \vec{A}$ 方向 $\mu_{\max} = |\text{rot} \vec{A}|$ (所沿方向余弦即 $(\text{rot} \vec{A})_M$ 方向余弦)

($\vec{A}$ 在 M 处绕 $\vec{n}_0$) $\mu = \text{rot} \vec{A} \cdot \vec{n}_0 = \begin{vmatrix} \cos\alpha & \cos\beta & \cos\gamma \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$ $\vec{n}_0 = \{\cos\alpha, \cos\beta, \cos\gamma\}$

$\text{rot} \vec{A} = 0$ 无旋场

通量与散度 (空间曲面)

通量 $\iint_S \vec{A} \cdot d\vec{S}$ 散度 $\text{div} \vec{A} = \lim_{S \rightarrow 0} \frac{\oiint_S \vec{A} \cdot d\vec{S}}{V} = \lim_{S \rightarrow 0} \frac{\iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dv}{V} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$ 流量密度 $\begin{cases} > 0 \text{ 源} \\ < 0 \text{ 汇} \\ = 0 \text{ 无源场} \end{cases}$

推论 ① $\text{div} \vec{A} = 0 \Rightarrow \oiint_S \vec{A} \cdot d\vec{S} = 0$

$\text{grad}(u) = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right)$

② 某处 $\text{div} \vec{A} \neq 0$ / 不恒为 0 $\Rightarrow \oiint_S \vec{A} \cdot d\vec{S}$ 处处等



小技巧

首先考虑对称性

$$\int_a^b dy \int_y^b f(x) dx = \int_a^b dx \int_a^x f(x) dy = \int_a^b (b-a) f(x) dx$$

r' 可由 r 的方程消元

关于积分方法的选择

$$\int_a^b dy \int_y^b f(x) f(y) dx = \frac{1}{2} \left[\int_a^b f(x) dx \right]^2$$

用方程整体消元

二重积分 = 平面投影 / 板 / 板

$$\oint_C x dy - y dx = \iint_D dx dy = 2S$$

三重积分 = 投影 (由 2 交点, 曲与平围成)

截面 (但且 $A(2)$ 好表示)

柱 ($6xy$ 为圆)

球 (曲曲交) + 一般

原函数

$$\nabla \left(\frac{1}{2} x^2 + \frac{1}{2} y^2 + \frac{1}{2} z^2 \right) = (x, y, z)$$

$$y z dx + x z dy + x y dz = d(xyz)$$

$$x^2 + y^2 + z^2 = R^2$$

$$S(R) = \iint_D \sqrt{1 + z_x^2 + z_y^2} = \iint_D \frac{R}{\sqrt{R^2 - x^2 - y^2}} dx dy$$

第二类曲面 =

$$\iint_S \vec{A} \cdot \vec{n} ds \rightarrow \iint_D R(x, y, z) d\sigma$$

第一类曲线 = $y'(x) / x'(t) / y'(t) / r'(t)$

第二类曲线 = 路径无关 (验证 / 凑原函数)

$\int P dx + Q dy + R dz$ 参数 $\rightarrow$ 第一类

曲面积分若投影为曲线 (非=维)
则积分值为 0

$$\iint_S (P dy dz + Q dz dx + R dx dy) = \iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dV$$

$$R^2 \phi = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) d\sigma \quad \text{补 / 循环常数}$$

$$R^3 \phi = \iint_D \begin{vmatrix} \frac{dy dz}{\partial x} & \frac{dz dx}{\partial y} & \frac{dx dy}{\partial z} \\ P & Q & R \end{vmatrix} d\sigma$$

讨论定义域

特殊积分

$$\textcircled{1} \int_0^{+\infty} \frac{t^2}{1+t^4} dt \xrightarrow{u=\frac{1}{t}} \int_{+\infty}^0 \frac{\frac{1}{u^2}}{1+\frac{1}{u^4}} d\left(\frac{1}{u}\right) = \int_0^{+\infty} \frac{1}{u^2+1} du = \frac{1}{2} \int_0^{+\infty} \frac{t^2+1}{t^4+1} dt$$

$$\text{倒变换} \int_{-\infty}^{+\infty} \frac{1}{x^4 \sqrt{ax^2+bx+c}} dx \quad \text{令 } x = \frac{1}{t}$$

$$\int_0^{+\infty} \frac{Ht^2}{Ht^4+1} dt = \int_0^{+\infty} \frac{Ht^2}{t^4+\frac{1}{H}} dt = \int_0^{+\infty} \frac{d(t-\frac{1}{t})}{(t-\frac{1}{t})^2+2} \quad x=t-\frac{1}{t} \quad \int_{-\infty}^{+\infty} \frac{dx}{x^2+2} = \frac{\sqrt{2}}{2} \arctan \frac{x}{\sqrt{2}} \Big|_{-\infty}^{+\infty} = \frac{\sqrt{2}}{2} \pi$$

$$\textcircled{2} \int \frac{x^2}{(1+x^2)^2} dx = \frac{1}{2} \int \frac{x^2}{(1+x^2)^2} d(x^2) = \frac{1}{2} \int x^2 d\left(\frac{1}{1+x^2}\right) = -\frac{x^3}{2(1+x^2)} + \frac{3}{2} \int \frac{x^2}{1+x^2} dx$$

$$= \int \left(1 - \frac{1}{1+x^2} \right) dx$$

$$\textcircled{3} \int \frac{1}{\sqrt[3]{(x+1)^2(x-1)^4}} dx = \int \frac{1}{(x-1)(x+1)\sqrt[3]{x+1}} dx$$



积分公式

$$\int \sqrt{a^2+x^2} dx = \frac{1}{2} \left(x \sqrt{a^2+x^2} + a^2 \ln \left| x + \sqrt{a^2+x^2} \right| \right) + C$$

$$\int \sqrt{x^2-a^2} dx = \frac{1}{2} \left(x \sqrt{x^2-a^2} - a^2 \ln \left| x + \sqrt{x^2-a^2} \right| \right) + C$$

$$\int \sqrt{a^2-x^2} dx = \frac{1}{2} \left(x \sqrt{a^2-x^2} + a^2 \arcsin \frac{x}{a} \right) + C$$

$$\int \frac{1}{\sqrt{a^2+x^2}} dx = \ln \left| \sqrt{x^2+a^2} + x \right| + C$$

$$\int \frac{1}{\sqrt{x^2-a^2}} dx = \ln \left| \sqrt{x^2-a^2} + x \right| + C$$

$$\int \frac{1}{\sqrt{a^2-x^2}} dx = \arcsin \frac{x}{a} + C$$

$$\int \frac{1}{a^2+x^2} dx = \frac{1}{a} \arctan \frac{x}{a} + C$$

$$\int \frac{1}{x^2-a^2} dx = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$$

$$\int \frac{1}{a^2-x^2} dx = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$$

$$\int \frac{1}{(a^2+x^2)^{\frac{3}{2}}} dx = \frac{x}{a^2 \sqrt{a^2+x^2}} + C$$

$$\int \tan x dx = -\ln |\cos x| + C$$

$$\int \cot x dx = \ln |\sin x| + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$\int \sec x dx = \ln |\sec x + \tan x| + C$$

$$\int \csc x dx = \ln |\csc x - \cot x| + C$$

$$\int \sec^3 x dx = \frac{1}{2} (\sec x \tan x + \ln |\sec x + \tan x|)$$

$$\sqrt{a^2+x^2} \xrightarrow{x=a \tan t} a \sec t$$

$$\sqrt{x^2-a^2} \xrightarrow{x=a \sec t} a \tan t$$

$$\int \frac{x^2}{\sqrt{a^2-x^2}} dx = \frac{1}{2} \left(a^2 \arcsin \frac{x}{a} - x \sqrt{a^2-x^2} \right) + C$$

$$\int \frac{x}{\sqrt{1+x^2}} dx = \sqrt{1+x^2} + C$$

$$\int \frac{1}{\sqrt{a-x}} dx = -2\sqrt{a-x} + C$$

$$\int \frac{x}{\sqrt{1-x^2}} dx = -\sqrt{1-x^2} + C$$

$$\int \frac{1}{a-x} dx = -\ln |a-x| + C$$

$$\int \sqrt{x} dx = \frac{2}{3} x^{\frac{3}{2}} + C$$

$$\int_0^{\pi} \sin^2 \theta d\theta = \frac{\pi}{2}$$

$$\int_0^{\pi} \sin \theta d\theta = 2 \quad \int_0^{\frac{\pi}{2}} \sin \theta d\theta = 1$$

$$\int \sin^2 \theta d\theta = \frac{1}{2} \theta - \frac{1}{2} \sin \theta \cos \theta + C$$

曲线

双曲线 $(x^2+y^2)^2 = a^2(x^2-y^2)$
 $r^2 = a^2 \cos 2\varphi$

螺旋线 $x = a \cos t \quad y = a \sin t \quad z = \frac{h}{2\pi} t$

星形线 $x = a \cos^3 t \quad y = b \sin^3 t$

$$(\tan x)' = \frac{1}{\cos^2 x} = \sec^2 x$$

$$(\cot x)' = \frac{1}{\sin^2 x} = -\csc^2 x$$

$$\int e^x \sin x dx = \frac{e^x}{2} (\sin x - \cos x) + C$$

$$x \sim e^x + 1 \sim \ln(x+1) \sim \frac{1}{2} [(1+x)^2 - 1] \sim \arcsin x \sim \arctan x \sim \sin x \sim \tan x$$

$$\int_0^{2\pi} \sin^{2n} x dx = 2 \int_0^{\pi} \sin^{2n} x dx = 4 \int_0^{\frac{\pi}{2}} \sin^{2n} x dx$$

$$\int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$$

$$\int_0^{\pi} \sin x \cos x dx = \frac{\pi}{2}$$

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\cos x \sin x}{1+e^x} dx$$

$$\int \frac{dx}{1 \pm \cos x} \quad \frac{1}{2} \tan \frac{x}{2} = t$$

$$\int \tan^5 x dx = \int \tan x (\sec^2 x - 1)^2 dx$$

